

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack platform (React front end, Node.js/MongoDB backend) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Owned backend reliability and correctness under load: added integration tests, monitored production logs and debugged tenant-specific incidents, hardened auth and rate limiting, and remediated **20+ VAPT findings**.
- Owned the eTriage backend end-to-end as primary technical POC, wrote knowledge-transfer documentation onboarding a second developer in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment for AI-assisted development.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and shipped **Underdogs Fitness** (Website) end-to-end, a production MERN platform (backend to product surface) with Stripe + manual-cash payments, cron automation, and multi-branch support; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a full-stack Python/FastAPI service turning open-ended queries into structured entity tables via typed planning, **deterministic extraction** with LLM fallback, output verification, and **cell-level provenance**; hardened with **150+ tests**.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built an evaluation-improvement agent that scores changes against multiple reference systems and commits only improvements to **Git** with report cards and **anti-Goodharting guardrails**, keeping evaluation auditable and hard to game.

Refusal Decay – LLM Safety / Mechanistic Interpretability (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Studied how prefilling attacks weaken refusal in Llama-3.1-8B-Instruct, extracting the residual-stream refusal direction via difference-in-means (refusal **0.92** → **0.32** at $k=3$) with causal patching, bootstrap CIs, and McNemar's test.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, JavaScript, TypeScript, C++, Java, SQL, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: FastAPI, Node.js, Express, React, Next.js, Git, GitHub Actions, REST APIs

LLM & Evaluation: LLM agents, LLM evaluation (anti-Goodharting), RAG, provenance/verification, PyTorch

Testing / Correctness: pytest, integration testing, deterministic extraction, CI/CD, code review, debugging

Cloud / Platforms: AWS, GCP, Azure, Docker, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.